

```
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21fc005b pa 0x07f22000 perm 0x5b
va 0x1000 pte 0x21fc7c1b pa 0x07f1f000 perm 0x1b
va 0x2000 pte 0x21fc7817 pa 0x07f1e000 perm 0x17
va 0x3000 pte 0x21fc7407 pa 0x07f1d000 perm 0x7
va 0x4000 pte 0x21fc70d7 pa 0x07f1c000 perm 0xd7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x6 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x6 perm 0x0
va 0x3fffff6000 pte 0x0 pa 0x0 perm 0x0
va 0x3fffff7000 pte 0x0 pa 0x0 perm 0x0
va 0x3fffff8000 pte 0x0 pa 0x0 perm 0x0
va 0x3fffff9000 pte 0x0 pa 0x0 perm 0x0
va 0x3fffffa000 pte 0x0 pa 0x0 perm 0x0
va 0x3fffffb000 pte 0x0 pa 0x0 perm 0x0
va 0x3fffffc000 pte 0x0 pa 0x0 perm 0x0
va 0x3fffffd000 pte 0x0 pa 0x0 perm 0x0
va 0x3fffffe000 pte 0x21fd08c7 pa 0x07f42000 perm 0xc7
va 0x3ffffff000 pte 0x2000184b pa 0x00000000 perm 0x4b
print_pgtbl: OK
vmprint_test starting
page table 0x0000000087f25000: pte 0x0000000021fc001 pa 0x0000000087f21000
..0x0000000000000000: pte 0x0000000021fc001 pa 0x0000000087f20000
..0x0000000000000006: pte 0x0000000021fc085b pa 0x0000000087f22000 R x U
..0x0000000000000100: pte 0x0000000021fc7c1b pa 0x0000000087f1f000 R x U
..0x0000000000000200: pte 0x0000000021fc7817 pa 0x0000000087f1e000 R W U
..0x0000000000000300: pte 0x0000000021fc7407 pa 0x0000000087f1d000 R W
..0x0000000000000400: pte 0x0000000021fc70d7 pa 0x0000000087f1c000 R W U
0x0000003fc0000000: pte 0x0000000021fc90d1 pa 0x0000000087f20000
..0x0000003ffffe0000: pte 0x0000000021fd08c7 pa 0x0000000087f42000 R W
..0x0000003fffff0000: pte 0x000000002000184b pa 0x0000000000000000 R x
vmprint_test: done
pgaccess_test starting
pgtbltest: pgaccess_test failed: incorrect access bits set, pid=3
$
```

```
hart 1 starting
hart 2 starting
init: starting sh
$ pgtbltest
print_ptbl starting
va 0x0 pte 0x21FD05B pa 0x87F7E000 perm 0x5B
va 0x1000 pte 0x21FDEC1B pa 0x87F7B000 perm 0x1B
va 0x2000 pte 0x21FDE817 pa 0x87F7A000 perm 0x17
va 0x3000 pte 0x21FDE407 pa 0x87F79000 perm 0x7
va 0x4000 pte 0x21FDE007 pa 0x87F78000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0xA000 pte 0x0 pa 0x0 perm 0x0
va 0xB000 pte 0x0 pa 0x0 perm 0x0
va 0xC000 pte 0x0 pa 0x0 perm 0x0
va 0xD000 pte 0x0 pa 0x0 perm 0x0
va 0xE000 pte 0x0 pa 0x0 perm 0x0
va 0xF000 pte 0x0 pa 0x0 perm 0x0
va 0x10000 pte 0x21FEC413 pa 0x87FB1000 perm 0x13
va 0x11000 pte 0x21FE7CC7 pa 0x87F9F000 perm 0xC7
va 0x12000 pte 0x2E001C4B pa 0x8B007000 perm 0x4B
print_ptbl: OK
vmprint_test starting
page table 0x000000007F81B00
0x0000000000000000: pte 0x000000021fd001 pa 0x000000007fd000
...0x0000000000000000: pte 0x000000021fd001 pa 0x000000007fc000
...0x0000000000000000: pte 0x000000021fd05b pa 0x000000007f7000 R X U
...0x0000000000001000: pte 0x000000021fd01b pa 0x000000007f7000 R X U
...0x0000000000002000: pte 0x000000021fd017 pa 0x000000007f7000 R W U
...0x0000000000003000: pte 0x000000021fd007 pa 0x000000007f7000 R W
...0x0000000000004000: pte 0x000000021fd0d7 pa 0x000000007f7000 R W U
0x0000003fc0000000: pte 0x000000021fe001 pa 0x000000007f8000
...0x0000003ffff00000: pte 0x000000021fd001 pa 0x000000007f7000
...0x0000003ffffd0000: pte 0x000000021fec413 pa 0x000000007fb1000 R U
...0x0000003ffffe0000: pte 0x000000021fe7cc7 pa 0x000000007f9f000 R W
...0x0000003fffff0000: pte 0x00000002e001c4b pa 0x000000000007000 R X
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
ugetpid_test: OK
ksuper_test: OK
pgtbltest: all tests succeeded
$
```

```
hart 1 starting
hart 2 starting
init: starting sh
$ pgtbltest
print_ptbl starting
va 0x0 pte 0x21FD05B pa 0x87F7E000 perm 0x5B
va 0x1000 pte 0x21FDEC1B pa 0x87F7B000 perm 0x1B
va 0x2000 pte 0x21FDE817 pa 0x87F7A000 perm 0x17
va 0x3000 pte 0x21FDE407 pa 0x87F79000 perm 0x7
va 0x4000 pte 0x21FDE007 pa 0x87F78000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0xA000 pte 0x0 pa 0x0 perm 0x0
va 0xB000 pte 0x0 pa 0x0 perm 0x0
va 0xC000 pte 0x0 pa 0x0 perm 0x0
va 0xD000 pte 0x0 pa 0x0 perm 0x0
va 0xE000 pte 0x0 pa 0x0 perm 0x0
va 0xF000 pte 0x0 pa 0x0 perm 0x0
va 0x10000 pte 0x21FEC413 pa 0x87FB1000 perm 0x13
va 0x11000 pte 0x21FE7CC7 pa 0x87F9F000 perm 0xC7
va 0x12000 pte 0x2E001C4B pa 0x80007000 perm 0x4B
print_ptbl: OK
vmprint_test starting
page table 0x000000007F81B00
0x0000000000000000: pte 0x000000021fd001 pa 0x000000007fd000
...0x0000000000000000: pte 0x000000021fd001 pa 0x000000007fc000
...0x0000000000000000: pte 0x000000021fd05b pa 0x000000007f7000 R X U
...0x0000000000001000: pte 0x000000021fd01b pa 0x000000007f7000 R X U
...0x0000000000002000: pte 0x000000021fd017 pa 0x000000007f7000 R W U
...0x0000000000003000: pte 0x000000021fd007 pa 0x000000007f7000 R W
...0x0000000000004000: pte 0x000000021fd0d7 pa 0x000000007f7000 R W U
0x0000003fc0000000: pte 0x000000021fe001 pa 0x000000007f8000
...0x0000003ffff00000: pte 0x000000021fd001 pa 0x000000007f7000
...0x0000003fffffd000: pte 0x000000021fec413 pa 0x000000007fb1000 R U
...0x0000003fffffe000: pte 0x000000021fe7cc7 pa 0x000000007f9f000 R W
...0x0000003fffff000: pte 0x00000002e001c4b pa 0x000000000007000 R X
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
ugetpid_test: OK
ksuper_test: OK
pgtbltest: all tests succeeded
$
```